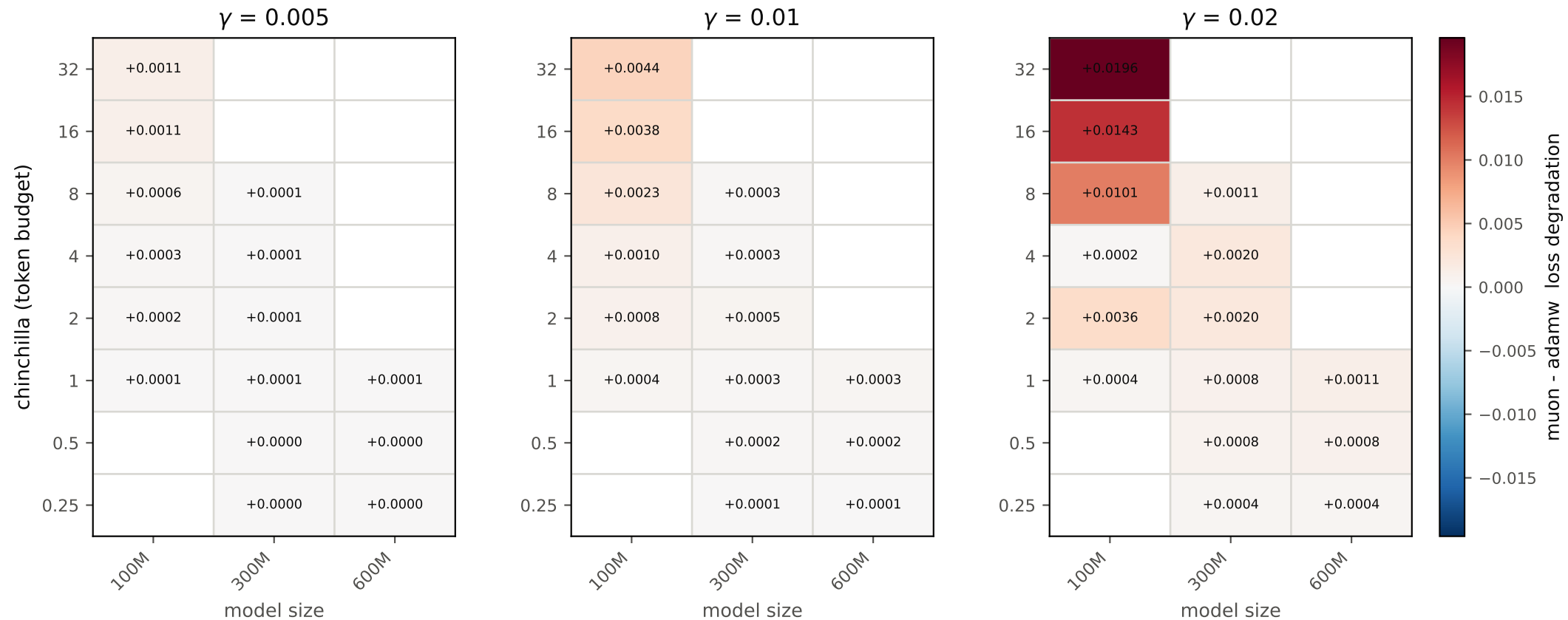


Degradation under Gaussian weight perturbation: muon vs adamw ($\gamma \leq 0.02$), 100M, 300M and 600M



cell = (perturbed - unperturbed) for muon minus the same for adamw | negative (blue) = muon degrades less